

Mode 1 — Square-BFP (centered / opt)

DP16 (8×4) → DP8 (8×4) → block exponents → DP8 (4×4) → centered aligned add-and-square

Step 1 — DP16 (8×4)

$$\text{DP16}(8 \times 4) = \sum_{i=0}^{15} a_i b_i$$

Step 2 — DP8 (8×4)

$$\text{DP16}(8 \times 4) = \underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8}(8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8}(8 \times 4)}$$

Step 3 — Block exponents (one exponent block per DP8 (8×4))

$$\left(\underbrace{\sum_{i=0}^7 a_i b_i}_{\text{DP8}(8 \times 4)}, \underbrace{\sum_{i=8}^{15} a_i b_i}_{\text{DP8}(8 \times 4)} \right) \longrightarrow \left(\underbrace{2^{-\delta_0} \sum_{i=0}^7 a_i b_i + 2^{-\delta_1} \sum_{i=8}^{15} a_i b_i}_M, E \right)$$

$E_0 = E_{a,0} + E_b$, $E_1 = E_{a,1} + E_b$; $E = \max(E_0, E_1)$, $\delta_k = E - E_k$; truncating shifts, on block sums only; $\text{DP16}(8 \times 4) = M \cdot 2^E$.

Step 4 — DP8 (4×4)

$$M = 2^{-\delta_0} \left(\underbrace{2^4 \sum_{i=0}^7 a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right) + 2^{-\delta_1} \left(\underbrace{2^4 \sum_{i=8}^{15} a_{H,i} b_i}_{\text{DP8}(4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i}_{\text{DP8}(4 \times 4)} \right)$$

Step 5 — Centered aligned add-and-square (subtract 8 from a_L ; each DP8 (4×4) = $\frac{1}{2} [\sum (x+b)^2 - \sum x^2 - \sum b^2 (+512 \text{ for } a_L)]$)

$$\begin{aligned} M = & 2^{-\delta_0} 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + b_i)^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 b_i^2 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_0} \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 - \sum_{i=0}^7 (a_{L,i} - 8)^2 - \sum_{i=0}^7 (b_i - 8)^2 + 512 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_1} 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} (a_{H,i} + b_i)^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} b_i^2 \right)}_{\text{DP8}(4 \times 4)} \\ & + 2^{-\delta_1} \underbrace{\frac{1}{2} \left(\sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 - \sum_{i=8}^{15} (a_{L,i} - 8)^2 - \sum_{i=8}^{15} (b_i - 8)^2 + 512 \right)}_{\text{DP8}(4 \times 4)} \end{aligned}$$

Step 6 — All components

$$\begin{aligned} \text{PE} &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 (a_{H,i} + b_i)^2 + \sum_{i=0}^7 ((a_{L,i} - 8) + b_i)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} (a_{H,i} + b_i)^2 + \sum_{i=8}^{15} ((a_{L,i} - 8) + b_i)^2 \right) \\ \alpha &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 a_{H,i}^2 + \sum_{i=0}^7 (a_{L,i} - 8)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} (a_{L,i} - 8)^2 \right) \\ \beta &= 2^{-\delta_0} \left(2^4 \sum_{i=0}^7 b_i^2 + \sum_{i=0}^7 (b_i - 8)^2 \right) + 2^{-\delta_1} \left(2^4 \sum_{i=8}^{15} b_i^2 + \sum_{i=8}^{15} (b_i - 8)^2 \right) \\ C &= 2^{-\delta_0} (0 + 512) + 2^{-\delta_1} (0 + 512) \\ M &= \frac{1}{2} (\text{PE} - \alpha - \beta + C), \quad E = \max(E_0, E_1) \end{aligned}$$